

CHAPTER 11

Function-Based Interventions

Education is the establishing of behavior which will be of advantage to the individual and to others at some future time.

—B. F. SKINNER (1953, p. 402)

Within a problem-solving framework, practitioners apply FBA procedures to (1) identify behaviors that interfere with students' functioning; (2) quantify present levels of interfering behaviors; (3) identify the interacting variables that contribute to occurrences of interfering behaviors; and (4) select, implement, and monitor interventions that weaken interfering behaviors and strengthen appropriate replacement behaviors (Steege & Pratt, 2012). Indirect, descriptive, and/or experimental FBA results thus drive intervention planning and implementation efforts. The synthesis of data using the Behavior-Analytic Problem Solving: Assessment Results recording form (Chapter 6) provides a "diagnosis" of the problem, but the ultimate goal of the problem-solving assessment process is to arrive at a solution. Would you be satisfied if you took your vehicle to a mechanic, received a diagnostic report that a head gasket was blown, and then had to drive off with a "Good luck!" and no repair? Or, what if you went to the hospital with severe abdominal pain, participated in a series of physical examinations, received a diagnosis of appendicitis, and then left with no treatment and only a few reassuring words from the doctor: "Yes, indeed, you have a problem. Your appendix is seriously infected. The pain will only get worse, and the appendix likely will burst and spread bacteria throughout your body. Tough luck!" You get the idea. Diagnosing a behavioral problem without offering a solution is far from best practice.

Conducting FBAs to identify the function(s) of interfering behaviors guides school-based teams toward solutions for referred problems. Research consistently shows that interventions informed by FBA results and matched to behavioral function are more effective

The Need for a Pliant* Game Plan

When we begin the FBA process, we usually start with the question, “What are we up against?” In other words, “What array of interacting variables serve to evoke, occasion, and reinforce interfering behaviors?” Assessment results then inform our development of a game plan. Our game plan is a set of tactics and strategies aimed both at neutralizing the variables controlling interfering behavior and strengthening the variables controlling appropriate behavior. Sometimes, though, we need to be flexible and adjust our game plan based on changing conditions.

Consider a football analogy. Prior to the game, the offensive coordinator identified a series of plays designed to expose and take advantage of the defensive alignments of the other team.

The team is in the huddle . . . a play is called in . . . the quarterback lines up behind the center . . . he observes the defensive alignment presented by the opponents . . . realizes that the called play feeds right into the strengths of the defense and is going to be a complete failure . . . being a flexible and pliable quarterback, he “reads” the defensive alignment . . . he identifies a weakness . . . he calls an audible . . . the team executes the play . . . and you know the rest of the story (another amazing comeback by the Patriots!).

So, yes, we may have a game plan for a function-based (function-focused) behavior support plan—but the plan needs to be flexible to take into account the dynamic relationships among variables that contribute to interfering behavior.

**Pliant:* Flexible and suitable for many uses.

than interventions developed arbitrarily (e.g., Campbell, 2003; Heyvaert, Maes, Van den Noortgate, Kuppens, & Onghena, 2012; Heyvaert, Saenen, Campbell, Maes, & Onghena, 2014; Repp, Felce, & Barton, 1988), and the behavior-analytic literature is replete with demonstrations of the effectiveness of individually tailored, function-based interventions for addressing interfering behaviors. These empirically supported interventions generally fall into three categories:

- Antecedent interventions to reduce the likelihood of interfering behaviors,
- Teaching and reinforcement strategies to strengthen replacement behaviors, and
- Consequence-based strategies to reduce occurrences of interfering behaviors.

In this chapter, we provide a brief review of evidence-based, function-based intervention strategies and offer a model for translating the Behavior-Analytic Problem Solving: Assessment Results form into a comprehensive and highly individualized behavior intervention plan.

From FBA to Behavior Support Plan

The Technical Assistance Center on Positive Behavioral Interventions and Supports, funded by the U.S. Department of Education’s Office of Special Education Programs, offers a wealth of resources to schools seeking to implement a multi-tiered approach to promoting social, emotional, and behavioral outcomes for all learners. In this chapter, we present one model for linking FBA results to behavior intervention plans. We also highly recommend browsing the technical assistance center (www.pbis.org) to learn about alternative models for developing function-based behavior support plans using a team-based approach.

ANTECEDENT INTERVENTIONS

Function-based antecedent interventions include a wide array of proactive strategies to prevent the occurrence of interfering behaviors by targeting the S^Ds and MOs. Consider, for example, a student who engages in bolting behavior to escape from the loud environment of the cafeteria. The student's bolting behavior is occasioned by the sight of a nearby exit (S^D), evoked by aversive auditory stimulation (MO), and maintained by individually mediated negative reinforcement. Function-based antecedent interventions therefore may include assigning the student to sit at a table located far from the exit with her back to the door (S^D modification) and providing her with a set of headphones to block the aversive noise (MO manipulation). Table 11.1 on the next page provides descriptions and examples of select evidence-based, function-based antecedent interventions.

Linking FBA Results to Antecedent Interventions

Completion of the Behavior-Analytic Problem Solving: Assessment Results recording form leads to the identification of three variables that may be targeted using antecedent interventions: context, motivational triggers, and sources of reinforcement.

Context

Identification of the context in which interfering behaviors occur helps the team modify the setting and/or make accommodations to minimize the likelihood of interfering behaviors. General strategies to consider include the following:

- *Modify the setting* to reduce the presence of stimuli that are aversive to the student. *Examples:* Cover fluorescent lights with fabric to reduce unpleasant flickering, install carpeting to dampen the noise in the classroom, position school staff in hallways to reduce the likelihood of bullying/teasing from peers.
- *Modify the student's participation in settings* that are associated with interfering behaviors. *Examples:* Minimize transitions and "downtime," adjust the relative amount of time spent in small-group versus large-group instruction, switch to an adaptive physical education class.

Motivational Triggers (MOs)

Identification of the motivational triggers for interfering behavior helps the team identify strategies to reduce (abolish) the value of the maintaining reinforcer and thereby reduce the momentary likelihood of interfering behavior. Remember, motivational triggers typically fall into two categories: the deprivation of reinforcers or the presentation of aversive stimulation. Therefore, antecedent interventions to reduce motivational triggers typically (1) prevent students from experiencing deprivation or (2) make the environment less aversive. General strategies to consider include the following:

TABLE 11.1. Selecting Function-Based Antecedent Interventions

Intervention	Description	Research Examples
<i>Antecedent interventions for interfering behaviors maintained by positive reinforcement</i>		
Noncontingent reinforcement (NCR)	Provide free access to edible, tangible, activity, social, or sensory reinforcers at regularly scheduled intervals of time (independent of student behavior) to prevent “deprivation” and reduce the motivation to engage in interfering behaviors maintained by access to those reinforcers.	Vollmer, Iwata, Zarcone, Smith, and Mazaleski (1993) delivered attention to participants at least once every 5 minutes, regardless of their behavior, to reduce attention-maintained self-injury. Hagopian, Crockett, Van Stone, DeLeon, and Bowman (2000) provided participants with brief access to reinforcing items every 30 seconds to 7 minutes to reduce aggression maintained by access to tangibles.
Stimulus control	Modify the environment to remove signals that indicate the availability of reinforcers for interfering behavior and/or introduce signals that indicate the unavailability of reinforcers for interfering behavior.	Maglieri, DeLeon, Rodriguez-Catter, and Sevin (2000) placed stickers on food containers to help the participant discriminate between permitted and prohibited food items and reduce occurrences of covert food stealing.
<i>Antecedent interventions for interfering behaviors maintained by negative reinforcement</i>		
Noncontingent escape (NCE)	Provide brief breaks at regularly scheduled intervals of time (independent of student behavior) to reduce the motivation to engage in escape-maintained interfering behaviors.	Vollmer, Marcus, Ringdahl, and Roane (1995) offered participants brief, scheduled breaks from instruction to reduce escape-maintained self-injury.
Stimulus pairing	Pair teachers with positive reinforcement to establish the value of social interaction as a reinforcer and reduce the motivation to escape from instructional sessions with the teacher.	Kelly, Axe, Allen, and Maguire (2015) conducted 2- to 4-minute pairing sessions (i.e., instructors provided high-quality attention and engaged students in preferred activities) prior to implementing discrete trial instructional sessions to reduce escape-maintained interfering behaviors.
Stimulus fading	Gradually expose students to stimuli (e.g., tasks) that evoke interfering behaviors to reduce the likelihood of those behaviors.	Pace, Iwata, Cowdery, Andree, and McIntyre (1993) eliminated and then gradually and systematically reintroduced instructions, while implementing escape extinction, to reduce participants’ levels of escape-maintained self-injury.

(continued)

TABLE 11.1. (continued)

Intervention	Description	Research Examples
High-probability request sequence/ task interspersal	Present a series of simple instructions with which compliance is likely prior to presenting an instruction that typically evokes escape-maintained behavior (e.g., “Give me a high five!”; “Make a silly face”; “Touch your toes”; “Line up for the bathroom”). Intersperse tasks that evoke escape-maintained behavior with less challenging/more preferred tasks.	Lee, Lylo, Vostal, and Hua (2012) presented students with a sequence of three high-preference math problems prior to presenting low-preference math problems to reduce latency to comply with academic task expectations.
Instructional/task modification	Modify the pace of instruction, task difficulty, type of prompts, error correction procedures, and/or other aversive aspects of the instructional environment to reduce the motivation to engage in escape-maintained interfering behaviors.	McComas, Hoch, Paone, and El-Roy (2000) used FBA data to identify which aspects of academic instruction evoked interfering behaviors for three students and then modified those variables (i.e., by offering instructional supports like calculators, offering choices for task sequencing, and providing nonrepeated tasks) to reduce escape-maintained destructive behavior.

- *Use noncontingent reinforcement* to reduce the student’s motivation to engage in interfering behaviors. *Examples:* Circulate around the classroom and engage with the student at least once every 5 minutes, schedule opportunities for peer interaction every 15 minutes during academic instruction, offer breaks from difficult academic assignments every 10 minutes.
- *Enrich the environment and modify the aversive aspects of the educational and social environment* that trigger interfering behaviors. *Examples:* Eliminate the use of physical prompting that is aversive to the student, utilize errorless teaching strategies to eliminate the need for corrective feedback that evokes interfering behavior, utilize incidental teaching procedures in lieu of discrete trial instruction that is aversive to the student, present instruction in the format of a game, assign a peer buddy to a student who dreads sitting alone at lunch.

Sources of Reinforcement (S^D s)

Identification of the persons, items, or locations that signal the availability of reinforcing consequences for interfering behaviors helps the team (1) *minimize* the signals that occasion interfering behaviors and (2) *maximize* the signals that occasion appropriate replacement behaviors. General strategies to consider include the following:

- Utilize a combination of *stimulus pairing and extinction* to establish a “new learning history” with individuals who historically provided reinforcement for interfering behaviors. *Example:* A teacher who functions as an S^D for escape-maintained behavior (1) increases his delivery of positive reinforcers and (2) stops permitting escape following interfering behavior.
- Modify the environment to *remove or minimize the presence of items that signal the availability of reinforcement*. *Examples:* Change classroom seating arrangements to move a student away from peers who function as S^Ds for attention-maintained behavior, move restricted items (e.g., iPads) out of sight and reach.
- *Introduce visual cues that signal the availability of reinforcement* for appropriate replacement behaviors. *Examples:* Utilize a token board and/or visual schedule to show expected behaviors and the availability of preferred items/activities.

TEACHING AND REINFORCEMENT-BASED INTERVENTIONS

Although antecedent interventions are highly effective for preventing occurrences of interfering behavior, *antecedent modifications yield only temporary effects*. For example, providing continuous social attention is likely to reduce occurrences of interfering behaviors maintained by access to teacher attention; however, there inevitably will be times when teachers need to divert their attention to other tasks or engage with other students. To achieve lasting behavior change, it is essential to teach students skills for coping with motivational triggers and alternative strategies to get their needs and wants met. For example, the student who engages in interfering behaviors to gain access to teacher attention may benefit from developing the skills to tolerate increasing durations of diverted/divided teacher attention, a stronger repertoire of independent academic and leisure skills, and more appropriate ways to recruit attention (e.g., hand raising, asking the teacher to look at completed work). Ultimately, teaching and reinforcing functional skills and appropriate replacement behaviors should be the primary focus of every behavior intervention plan.

Effective, function-based interventions that focus on strengthening appropriate behaviors generally involve:

- Delivering *direct, evidence-based instruction* in relevant communication, academic, executive, adaptive living, social, and/or self-management skills, and
- Providing *robust reinforcement for target skills/replacement behaviors* that effectively competes with the reinforcement maintaining interfering behaviors.

Functional communication training (FCT) is an evidence-based intervention that illustrates a method for blending both of these strategies into a single treatment package. To implement FCT, practitioners (1) identify a communication response that produces the same reinforcing consequences maintaining the student’s interfering behavior, (2) utilize effective instructional techniques and prompting strategies to teach the student to produce the target communication response, and (3) reinforce target communication responses by honoring the student’s requests (Tiger, Hanley, & Bruzek, 2008; Wacker et al., 1990). Depending on the function of the interfering behavior, a student may be taught to request access to atten-

tion (e.g., “Excuse me”; “Come play with me”; “Can we talk?”; “Look what I did!”), access to specific items/activities (e.g., “I want music, please”; “Can I have 1 more minute?”), escape from tasks/situations (e.g., “I need a break”; “I want to leave”), or removal of aversive stimuli (e.g., “Quiet, please”; “I don’t want that”). Although FCT is most appropriate when communication skill deficits contribute to occurrences of interfering behaviors, the basic conceptual framework—identifying, teaching, and reinforcing socially appropriate behaviors to replace interfering behaviors—is applicable to the treatment of most interfering behaviors. For example, an adolescent who engages in self-cutting to provide relief from unpleasant thoughts and emotions (automatic negative reinforcement) may benefit from an intervention that involves teaching and reinforcing alternative self-management strategies such as deep breathing and progressive muscle relaxation. Or, a middle school student who exhibits high levels of off-task behavior may benefit from an intervention that involves teaching self-monitoring skills and reinforcing on-task behavior.

Everything You Need to Know about FCT

To expand your understanding of FCT, we highly recommend the following book. It offers an outstanding blend of theory, applied research, methods, and examples.

Reichle, J., & Wacker, D. P. (2017). *Functional communication training for problem behavior*. New York: Guilford Press.

These practical guides and training resources also may be helpful:

Griffin, W., & AFIRM Team. (2017). *Functional communication training*. Chapel Hill: National Professional Development Center on Autism Spectrum Disorder, FPG Child Development Center, University of North Carolina. Retrieved from <http://afirm.fpg.unc.edu/functional-communication-training>.

Tiger, J. H., Hanley, G. P., & Bruzek, J. (2008). Functional communication training: A review and practical guide. *Behavior Analysis in Practice*, 1, 16–23.

Lastly, check out some of our own research*:

DeRosa, N. M., Fisher, W. W., & Steege, M. W. (2015). An evaluation of time in establishing operation on the effectiveness of functional communication training. *Journal of Applied Behavior Analysis*, 48, 115–130.

Shamlian, K. D., Fisher, W. W., Steege, M. W., Cavanaugh, B. M., Samour, K., & Querim, A. C. (2016). Evaluation of multiple schedules with naturally occurring and therapist-arranged discriminative stimuli following functional communication training. *Journal of Applied Behavior Analysis*, 49, 228–250.

Steege, M., Wacker, D., Cigrand, K., Berg, W., Novak, C., Reimers, T., et al. (1990). Use of negative reinforcement in the treatment of self-injurious behavior in children with severe multiple disabilities. *Journal of Applied Behavior Analysis*, 23, 459–467.

Wacker, D., Steege, M., Northup, J., Sasso, G., Berg, W., Reimers, T., et al. (1990). A component analysis of functional communication training across three topographies of severe behavior problems. *Journal of Applied Behavior Analysis*, 23(4), 417–429.

*OK, we admit it . . . blatant self-promotion—but we’re proud of our research. If you conduct a comprehensive review, then you will discover many studies demonstrating creative and innovative applications of FCT across diverse populations and settings.

Linking FBA Results to Teaching and Reinforcement-Based Strategies

Completion of the Behavior-Analytic Problem Solving: Assessment Results recording form leads to the identification of multiple variables that may be targeted using teaching and reinforcement-based strategies. Most relevant are the data obtained about skill delays/deficits and the reinforcing consequences maintaining interfering behavior.

Skill Delays/Deficits

As we discussed in Chapter 6, most interfering behaviors are directly related to weaknesses in communication, academic, executive, adaptive living, social, and/or self-management skills. Identification of relevant skill delays/deficits therefore helps the team identify target skills that may be strengthened to increase the likelihood of appropriate behaviors relative to the likelihood of interfering behaviors. Target skills and instructional methods are as unique and varied as the students we serve, so we offer the following only as illustrative examples of strategies to consider:

- Conduct *FCT* to strengthen socially appropriate communication responses that are functionally equivalent to the interfering behavior (e.g., Carr & Durand, 1985).
- Provide *direct instruction* in reading, writing, and/or mathematics skills (Stockard, Wood, Coughlin, & Rasplika Khoury, 2018).
- Implement an evidence-based *social-emotional skills curriculum* such as Skillstreaming (McGinnis, 2011a, 2011b) or Strong Kids (Carrizales-Engelmann, Feuerborn, Gueldner, & Tran, 2016).
- Utilize *modeling, rehearsal, and reinforcement strategies* to teach specific social, executive, or self-management skills (e.g., Peters & Thompson, 2015).
- Use *task analyses, prompting, chaining, and reinforcement procedures* to teach adaptive living skills (e.g., Stokes, Cameron, Dorsey, & Fleming, 2004).

Reinforcing Consequences

Identification of the reinforcing consequences maintaining interfering behaviors helps the team develop effective reinforcement programs to strengthen appropriate replacement behaviors. When we determine the reinforcing consequence(s) maintaining interfering behavior, we have, in effect, identified potential reinforcers for strengthening appropriate behavior. For example, if FBA results indicate that an interfering behavior is reinforced by access to social attention (socially mediated positive reinforcement), then it may be effective

Resources: Social–Emotional Learning

Check out Merrell's Strong Kids. These manualized programs use a cognitive-behavioral approach to teach social and emotional skills that promote resilience and mitigate risk in children and adolescents.

See the following websites:

- <https://strongkidsresources.com>
- <http://strongkids.uoregon.edu/research.html>

Self-Management 101

B. F. Skinner was an extraordinarily productive individual, in large part because of his engagement in self-management (Epstein, 1997). Self-management refers to the *personal* application of behavior-analytic strategies to produce behavior change. Self-management may be used break bad habits and acquire new ones, accomplish difficult tasks, and achieve personal goals (Cooper et al., 2007). Critical components of self-management interventions include:

- Antecedent modification: Self-modification of S^ps or MOs to increase the probability of desired behavior and reduce the probability of interfering behavior.
- Self-monitoring: Self-observation and behavior recording.
- Self-administered consequences: Self-application of consequences that are analogous to positive and negative reinforcement and positive and negative punishment.

to design an intervention in which social attention is provided contingent on appropriate behavior. Remember the matching law? The matching law predicts that minimizing social attention following interfering behavior, while providing robust social reinforcement for appropriate behavior, will shift responding to favor the appropriate behavior.

Reinforcement-based strategies include the following:

- *Differential reinforcement of alternative behavior (DRA)* involves delivering functional reinforcers for any socially appropriate replacement behavior while withholding (or at least minimizing) reinforcement for the interfering behavior. *Example:* Reinforce hand raising with social attention to replace calling-out behavior maintained by social attention.
- *Differential reinforcement of incompatible behavior (DRI)* involves delivering functional reinforcers for a replacement behavior that is physically incompatible with the interfering behavior while withholding (or at least minimizing) reinforcement for the interfering behavior. *Example:* Prompt and reinforce two-handed nose blowing using a tissue as a replacement for nose-picking behavior.
- *Differential reinforcement of communication behavior (DRC)* involves delivering functional reinforcers for a socially appropriate communication response while withholding (or at least minimizing) reinforcement for the interfering behavior. *Example:* Reinforce requests for a “break” as a replacement for vocal opposition maintained by escape from tasks.
- *Differential reinforcement of other behavior (DRO)* involves delivering functional reinforcers contingent on the absence of the interfering behavior. *Example:* Deliver social praise for the absence of attention-maintained screaming behavior at regularly scheduled intervals of time.
- *Contingency contracting* involves developing behavioral contracts that specify replacement behavior expectations and associated reinforcing consequences (e.g., Miller & Kelley, 1994). *Example:* Develop and implement a behavioral contract that describes requirements for work completion in order to earn breaks.
- *Token reinforcement plans* involve delivering tokens, exchangeable for functional reinforcers, contingent on the occurrence of a target replacement behavior and/or the nonoccurrence of interfering behavior. *Example:* Using a token board, deliver one token for every third math problem completed, and allow the student to take a break after earning 10 tokens.

To design a maximally effective reinforcement program, it is helpful to remember that both interfering behaviors and replacement behaviors are influenced by the same variables: MOs, S^Ds, and reinforcing consequences. Therefore, we encourage you to consider the following questions and strategies to increase the likelihood that your reinforcement procedures will be effective:

- **Questions:** Are the reinforcers for replacement behaviors valuable in the moment? Is motivation for the reinforcer strong?
 - **Strategies:** Conduct frequent preference assessments, control access to and vary the available reinforcers to prevent satiation, and give students “free samples” to establish motivation.
- **Questions:** Are relevant S^Ds present to signal the availability of reinforcement for replacement behaviors? Does the student know what to do to get what he or she wants?
 - **Strategies:** Use visual and verbal cues to prompt replacement behaviors and remind students about available reinforcers.
- **Questions:** Are reinforcers for replacement behaviors delivered consistently and immediately? Are the reinforcers of sufficient quality and magnitude given the response effort required? Is the reinforcement plan adequately robust?
 - **Strategies:** Analyze the parameters of reinforcement maintaining interfering behaviors and design reinforcement for the replacement behaviors to “compete” in terms of schedule, quality, magnitude, and delay. The goal is to ensure that replacement behaviors are more efficient and effective than the interfering behaviors!

CONSEQUENCE-BASED STRATEGIES TO REDUCE INTERFERING BEHAVIOR

Relative to reinforcing consequences, FBA results most obviously tell us *what not to do*. Remember, reinforcing consequences are the outcomes that follow occurrences of interfering behavior and strengthen its future probability. When our goal is to *weaken* the future probability of interfering behaviors, it certainly makes sense to stop reinforcing them! Extinction is the technical term for the procedure of withholding reinforcement of a response, and it results in a gradual reduction in responding. When interfering behaviors no longer “work,” they gradually fade away.

SOMETIMES THINGS GET WORSE BEFORE THEY GET BETTER . . .

Extinction is a powerful tool for reducing occurrences of interfering behavior. As with any powerful tool, though, caution must be taken to use extinction safely and effectively. Side effects, such as an extinction burst and aggressive/emotional behavior, may occur (Lerman, Iwata, & Wallace, 1999). The term *extinction burst* refers to temporary increases in the frequency, duration, intensity, and/or variability of behavior when reinforcement is discontinued. Imagine, for example, that parents no longer give in to their child’s demands when tantrums occur. The initial outcome of this change in parenting tactics may be increased rates of longer, and definitely more dramatic, tantrums. Although this “burst” will not sustain if the parents hold their ground, the parents certainly need be prepared to weather the storm.

A REAL-LIFE CAUTIONARY TALE . . .

In the late 1970s, I (Steege) was a houseparent in a group home serving children with developmental disabilities. Kent, age 17, had very limited communication and independent leisure skills. He also engaged in inappropriate kissing behavior that we hypothesized was reinforced by social attention (socially mediated positive reinforcement). That is, Kent would sneak up on people (even folks he did not know) and plant sloppy kisses on their faces, which needless to say produced quite a reaction! After attending a 1-day workshop on “behavior management” in which planned ignoring was enthusiastically discussed, my wife, Lisa, and I decided to ignore Kent each time he displayed this kissing behavior. One evening we were hanging in the family room watching a movie on TV. Kent snuck up from behind and kissed me on the cheek. I stoically offered no response. I was a rock. I froze all facial features and held my body rigid for 30 seconds. A few minutes later, Kent kissed me again. And I continued to use my “I am a statue” planned ignoring strategy. This pattern repeated itself several more times over the next 5 minutes, and then Kent unleashed a full-fledged “shock and awe” wave of unrelenting kissing. The kisses were steadily increasing in duration and intensity. Moreover, the time between kisses (a more fancy term would be “interresponse time”) was diminishing to only a few seconds. I did not know what to do. Now at this point I need to admit that I am a counter. Yes, I like to count things . . . I’ve got this thing for data . . . and when the kissing behavior had reached a frequency of 85 that evening, I was starting to get a bit frustrated. After another barrage of kissing, I looked at Lisa and pleaded for help. With an expression of heartfelt sympathy and compassion, she leaned over and empathically whispered, “You’re screwed.” So, when the kissing behavior surpassed 104, I jumped up from the couch, looked Kent squarely in the eyes, and yelled “Stop! That’s it! No more kissing.” With a mocking smile and a definite air of accomplishment, Kent sat on the couch and relaxed into a state of triumphant victory.

NOW, THAT’S A RESPONSE BURST!

So what went wrong with my intervention for Kent’s kissing behavior? Several things, of course. My movie-watching behavior led to a temporary deprivation of social interaction for Kent (an MO increasing the value of attention and the probability of kissing behavior as a way to obtain it). Kent’s deficits in independent leisure and communication skills meant that (1) he was bored and (2) he did not really know any better way to get my attention. I also made the mistake of *ignoring Kent*, rather than *Kent’s kissing behavior*. This is a subtle, but critical, point. By ignoring Kent entirely, I increased his motivation for obtaining attention of any form.

So what should I have done? Several things, of course. I should have recognized that Kent was bored, engaged him in a functional activity, and reinforced his participation. I should have conducted a reinforcer assessment to identify effective and powerful reinforcers for participation that would compete with the social attention he received following kissing behavior. I should have provided noncontingent reinforcement by offering social feedback intermittently while he participated in a functional activity. And, if kissing still occurred, I should have briefly ignored this behavior and then immediately redirected Kent to the activity so that I could provide social reinforcement for participation. I also should have used FCT strategies to teach appropriate communication behavior as a replacement for inappropriate kissing. I’m sure there’s even more I could have done, but I think it gives a clear picture of why a powerful behavior change strategy—extinction—worked against me.

THE MORAL OF THE STORY

Before proceeding with the use of extinction, make sure that you:

- Can safely withstand and tolerate an extinction burst, and
- Implement antecedent interventions and reinforcement-based procedures concurrent with extinction to reduce the probability of undesirable side effects and increase the likelihood of enduring behavior change.

Linking FBA Results to Consequence-Based Strategies to Reduce Interfering Behavior

Identification of the function of interfering behavior helps teams develop appropriately matched extinction procedures or strategies that minimize the reinforcement for interfering behavior relative to the reinforcement for appropriate replacement behavior. The key is to withhold or minimize the *functional* reinforcing consequence that maintains occurrences of interfering behavior. Table 11.2 summarizes function-based variations of extinction (Iwata, Pace, Cowdery, & Miltenberger, 1994).

An Alternative to Extinction

Sometimes extinction in its “pure” form cannot be implemented. For example, if a student engages in severe physical aggression to escape from nonpreferred academic tasks, it may not be safe or feasible for school staff to implement escape extinction by continuing to deliver instructions when aggression occurs. In these situations, we recommend applying the matching law by minimizing reinforcement for interfering behavior relative to reinforcement for appropriate replacement behavior. The basic strategy is to:

- Minimize the rate, quality, magnitude, and immediacy of reinforcement for the interfering behavior, and
- Maximize the rate, quality, magnitude, and immediacy of reinforcement for the replacement behavior.

For the student who engages in escape-maintained aggression, this strategy could involve:

- Returning to the initial task as soon as feasible, expecting completion of the initial task before moving forward in the activity schedule, and withholding access to preferred items/activities/social interactions during the escape period; and
- Reinforcing a low-effort communication response (e.g., touching a break card) with consistent, immediate access to a break.

BEHAVIOR-ANALYTIC PROBLEM SOLVING: FUNCTION-FOCUSED INTERVENTION

In Chapter 6, we presented the Behavior-Analytic Problem Solving: Assessment Results recording form to synthesize FBA data and facilitate an understanding of the variables that influence interfering behavior. By completing this form, an evaluator is effectively “diagnosing” the problem and delineating the multiple variables that need to be addressed to arrive at a solution to the problem. At this point, we offer an additional tool—the Behavior-Analytic Problem Solving: Function-Focused Intervention planning form—to translate the analysis of the interfering behavior into an individually tailored behavior intervention plan. This planning tool is displayed in Form 11.1 on page 211.

Using the Behavior-Analytic Problem Solving: Function-Focused Intervention Planning Tool

You will notice that the format of the Behavior-Analytic Problem Solving: Function-Focused Intervention planning tool generally mirrors the format of the Behavior-Analytic Problem

TABLE 11.2. Function-Based Extinction Procedures

Reinforcing consequence for interfering behavior	Function-based extinction procedure	Clinical example	Research example
Access to social attention	Withhold social attention following the interfering behavior.	Adam emits silly noises to get a reaction (a reprimand) from his teacher, so the teacher stops responding to silly behavior with reprimands.	Fisher, DeLeon, Rodriguez-Catter, and Keeney (2004) ignored occurrences of attention-maintained destructive behavior.
Access to tangibles/activities	Withhold access to tangibles/activities following the interfering behavior.	Julia engages in tantrums to get access to preferred toys, so her paraprofessional discontinues the practice of giving toys to “calm her down” when she tantrums.	Rispoli, Camargo, Machalicek, Lang, and Sigafooset (2014) withheld access to preferred activities and routines (e.g., lining up trains and repeatedly completing the same puzzle) contingent on interfering behaviors maintained by access to those routines.
Escape from tasks	Do not terminate the task following interfering behavior.	Mary engages in screaming to escape task demands, so the teacher continues delivering instruction using a least-to-most prompting hierarchy when screaming occurs.	Iwata, Pace, Kalsher, Cowdery, and Cataldo (1990) continued presenting tasks and used physical guidance to ensure task completion contingent on escape-maintained self-injury.
Escape from social interactions	Do not terminate social interactions following interfering behavior.	Phil yells “Go away” when teachers attempt to engage him in conversation, so teachers continue engaging with him when he yells.	Harper, Iwata, and Camp (2013) remained in close proximity and continued to deliver prompts contingent on occurrences of aggression maintained by escape from social interactions.
Automatic reinforcement	Block or mask the sensory consequences for interfering behavior.	Delia engages in skin picking (on the arms) maintained by automatic reinforcement, so her parents send her to school wearing a long-sleeved body suit that blocks the sensory consequence.	Roscoe, Iwata, and Goh (1998) used gloves and/or protective sleeves to attenuate the sensory consequences maintaining self-injurious behaviors.

Solving: Assessment Results recording form. This tool is intended to be completed in collaboration with the student's school-based team, after reviewing FBA assessment results, to generate strategies for addressing each of the individual and environmental variables that contribute to the interfering behavior. In the following subsections of this chapter, we review each component of the Behavior-Analytic Problem Solving: Function-Focused Intervention planning form and provide examples of relevant intervention strategies. (*Note:* The examples are for illustrative purposes only and do not reflect the comprehensive range of evidence-based intervention options that may be considered.)

Context

PROCESS

Given information about the contexts in which interfering behavior is most probable, identify strategies for modifying those contexts and/or the student's participation in those contexts.

EXAMPLE

FBA results indicated that Mabel's interfering behaviors were most probable during unstructured "downtime" in the classroom. To address this context variable, the team proposed (1) reducing the number and duration of unstructured periods of time and (2) providing Mabel with a menu of activity options from which to select during unstructured periods of time.

Personal Characteristics

PROCESS

Given information about personal characteristics that contribute to interfering behavior, identify relevant strategies to address those characteristics.

EXAMPLE

FBA results indicated that Alberto, a student with ASD, maintains poor dental hygiene and likely has untreated cavities. It was hypothesized that tooth pain evoked self-injurious behavior maintained by automatic negative reinforcement. The team therefore referred Alberto to a pediatric dentist with experience serving children with ASD and added an IEP goal to increase "compliance with dental hygiene routines" using desensitization and reinforcement strategies.

Skill Delays/Deficits

PROCESS

Given information about skill delays/deficits that contribute to interfering behavior, identify evidence-based instructional strategies to mitigate weaknesses.

EXAMPLE

FBA results indicated that Tasha exhibited significant deficits in reading fluency, which helped the team understand why she engaged in escape-maintained behaviors in the context of her regular education literacy class. The team then decided to refer Tasha to the school's reading specialist for intensive, individualized reading fluency intervention.

Motivational Triggers (MOs)

PROCESS

Given information about the motivational triggers for interfering behavior, identify strategies to weaken (abolish) that motivation.

EXAMPLE

FBA results indicated that Mason's argumentative behaviors were evoked by the presentation of academic tasks that required extensive writing (by hand). To reduce the aversive nature of academic tasks that required writing and reduce the likelihood of argumentative behaviors, the team provided Mason with access to a scribe and/or laptop as appropriate.

Sources of Reinforcement (S^Ds)

PROCESS

Given information about the stimuli that signal the availability of reinforcement for interfering behaviors, identify strategies to eliminate or modify these signals.

EXAMPLE

For Sandy, FBA results suggested that an open door to the principal's office signaled the availability of reinforcement (both social attention from the principal and access to her candy bowl) and occasioned bolting behavior during transitions from the classroom to the restroom. The team decided to ask Sandy's one-on-one support staff to call the office and request that they close the door to the principal's office prior to each transition.

Behavior

PROCESS

Identify the target interfering behavior and a logical, socially appropriate replacement behavior.

EXAMPLE

Jethro often bolted from the classroom to sit in the school lobby and watch visitors enter and exit the building. Jethro communicated primarily using the Picture Exchange Communica-

tion System (PECS); therefore, the team decided to teach him to request access to the school lobby by exchanging a picture symbol.

Reinforcing Consequences

PROCESS

Given information about the reinforcing consequences maintaining interfering behavior, identify strategies for (1) minimizing reinforcement of the interfering behavior and (2) maximizing reinforcement of the appropriate replacement behavior.

EXAMPLE

FBA results indicated that Barbara emitted loud, oppositional vocalizations (e.g., screeching, saying no, and swearing) to terminate intensive “table-time” instructional sessions. The team decided to place interfering behavior on extinction by continuing to present instructional prompts every 15 seconds when oppositional vocalizations occurred. Additionally, the team decided to teach Barbara to request a break from instructional sessions. Initially, to make break requests more probable than oppositional vocalizations, the team selected a low-effort communication response (touching a break card) and immediately reinforced every communication response with a 2-minute break on the beanbag. They also agreed to monitor Barbara’s behavioral data on a daily basis to determine when it was appropriate to *fade* the intervention by increasing the academic work expectations prior to honoring break requests.

Don’t Fall into the Trap!

When school-based teams are tasked with identifying solutions for behavioral “problems,” it is easy to fall prey to the trap of focusing on reactive interventions by asking questions like “What should we do when the interfering behavior occurs?” and “How should we respond?” Reactive interventions include strategies that are implemented *after* an interfering behavior occurs to reduce the probability of recurrence. From our perspective, the reduction or elimination of interfering behaviors is necessary, but hardly sufficient.

By conducting a comprehensive FBA and conceptualizing interfering behavior as the result of the confluence of individual, antecedent, and consequence variables, it becomes clear that effective problem solving also requires asking these questions: “What should we do *before* an interfering behavior occurs to prevent its occurrence?” and “What replacement behaviors should we teach and reinforce to make interfering behavior less likely?”

Interventions that address only one variable that contributes to interfering behavior are incomplete and unlikely to yield socially significant outcomes. Developing a behavior intervention plan that focuses exclusively on antecedent, teaching, and/or reinforcement-based procedures leaves teams without guidance on how to respond effectively (by maintaining safety and minimizing reinforcement) when interfering behaviors occur. And developing a behavior intervention plan that focuses exclusively on reactive procedures teaches a student *what not to do*, without teaching *what to do*. (Just imagine telling a student not to make mistakes when performing subtraction problems without teaching the student how to perform subtraction operations correctly!)

Maximally effective behavior intervention plans emphasize both (1) the acquisition and strengthening of socially appropriate skills and (2) the reduction of interfering behavior by addressing the comprehensive range of individual, antecedent, and consequence variables that contribute to the “problem.”

Perhaps You're Wondering . . .

So, what about punishment? Punishment refers to events that follow occurrences of interfering behavior and reduce the probability of its recurrence. Common punishment procedures used in schools include:

- Delivering reprimands/social disapproval (e.g., when a student hits, saying, “No thank you”).
- Positive practice (e.g., when a student runs in the hallway, requiring her to practice walking appropriately in the hallway five times).
- Restitution (e.g., when a student scribbles graffiti on a desk, requiring her to clean all the desks in the classroom).
- Time-out (e.g., when a student gets into an argument with a peer during a game of kickball, requiring him to “take the bench” for 5 minutes).
- Response cost (e.g., when a student engages in disrespectful behavior toward a teacher, she loses 5 minutes of recess).

Punishment procedures can be very effective and tend to promote rapid reductions in interfering behavior. However, punishment procedures are also highly susceptible to *overuse* and *misuse*. *Overuse* occurs because punishment often seems to work in the moment. For example, reprimands often immediately interrupt the interfering behavior. Remember, though, *consequences affect future behavior*. If a reprimand results in the immediate cessation of interfering behavior but yields no reductive effect on interfering behavior in the future, then the reprimand did not function as a punisher! *Misuse* occurs when school staff simply assume that a particular event will be an effective punisher for interfering behavior. Loss of recess privileges seems pretty aversive, right? It seems logical to assume that a student would be less likely to repeat a behavior that resulted in the loss of recess privileges . . . but that is a dangerous assumption. What if a student absolutely dreads recess? What if the student repeatedly experiences teasing at recess? For that student, recess may be an unpleasant part of the day that the student is highly motivated to avoid. Taking away that student's recess privileges contingent on interfering behavior may actually function as negative reinforcement, allowing the student to avoid aversive social circumstances and thereby increasing the future probability of interfering behavior. For another example of the misuse of punishment, see the box “A Word or Two about Time-Out.”

Given the potential benefits and risks of punishment, we advise school-based practitioners to follow the standards established by the BACB's (2014) *Professional and Ethical Compliance Code for Behavior Analysts* (Section 4.08), which require behavior analysts to:

- Recommend and implement reinforcement-based procedures rather than punishment whenever possible;
- Use punishment only in conjunction with reinforcement-based procedures to strengthen alternative behaviors; and
- Utilize punishment responsibly by (1) effectively training and supervising staff to implement the procedures, (2) closely monitoring the effectiveness of the procedures, (3) modifying procedures that are ineffective, and (4) fading the use of the procedures as quickly as possible.

Case Example of Function-Focused Planning

Figures 11.1 and 11.2 show completed versions of the Behavior-Analytic Problem Solving: Assessment Results recording form and the Behavior-Analytic Problem Solving: Function-Focused Intervention planning tool, respectively, to illustrate the approach of translating FBA data into an individually tailored, function-focused behavior intervention plan. Figure 11.1 summarizes the results of an FBA conducted with Mitch, a 16-year-old who had received multiple disciplinary referrals for oppositional and disruptive behaviors. Figure 11.2 presents the intervention recommendations that were generated by Mitch and his school-based team after a comprehensive review of the FBA results.

Personal Characteristics: Traits, sensitivities, beliefs, and/or values contributing to interfering behavior	Context: Probable contexts in which interfering behavior occurs	Skill Delays/Deficits: Communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior
- History of social anxiety - Sensitivity to negative peer evaluation - Belief that males cannot show weakness	General education classes that require high levels of oral participation	- Emotional regulation skill deficits - Below-average academic performance in all content areas
Motivational "Triggers" (Motivating Operations): Deprivation of physical or social wants and needs, which increases the value of reinforcers and the probability of interfering behavior Presentation of aversive biological/medical conditions, emotional states, thoughts, academic tasks, or social situations, which increases the value of reinforcers and the probability of interfering behavior		
- Direct instructions to respond orally to questions and/or participate in class discussions increase the value of escape from the classroom and trigger interfering behaviors - Unpleasant physiological arousal and worries about negative peer evaluation increase the value of arousal reduction and trigger interfering behaviors		
Sources of Reinforcement (Discriminative Stimuli): Persons, items, or locations that signal the availability of reinforcing consequences (i.e., source of reinforcement), thereby occasioning interfering behavior		
- Classroom teacher with a history of sending students to the principal's office for disruptive behavior - Sight/proximity of the exit door in the classroom		
Interfering Behavior(s):		
- Vocal opposition: Vocally refusing to comply with a teacher directive by saying no, arguing, and/or swearing - Leaving the area: Exiting the classroom without explicit teacher permission		
Reinforcing Consequence(s): Consequence(s) that strengthen the future probability of the interfering behavior	Parameters of Reinforcement: Describe the power (robustness) of reinforcing consequences	
- Escape from the classroom and associated expectations - Socially mediated negative reinforcement when directed by the teacher to the principal's office - Individually mediated negative reinforcement when independently leaving the area - Arousal reduction - Automatic negative reinforcement	- Teacher sends student to the principal's office after 30 seconds to 1 minute of intense swearing, and student escapes the classroom for about 15 minutes in a location where no peers are present to judge him - Student experiences immediate relief from negative thoughts and emotions	

FIGURE 11.1. Behavior-Analytic Problem Solving: Assessment Results for Mitch. Used with permission from the Margaret Murphy Centers for Children.

Personal Characteristics: Identify strategies to address traits, sensitivities, beliefs, and/or values contributing to interfering behavior Referral to school psychologist for cognitive-behavioral therapy to address social anxiety and related thoughts/emotions	Context: Identify strategies for modifying contexts and/or participation in contexts in which interfering behavior occurs Present option for participating in core instruction in the high school resource classroom	Skill Delays/Deficits: Identify strategies to mitigate communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior • Deliver specially designed instruction to address academic skill deficits • Teach emotional regulation skills such as thought stopping/reframing, deep breathing, and muscle relaxation
---	---	--

Motivational “Triggers” (Motivating Operations): Identify strategies to weaken (abolish) motivation for interfering behaviors by preventing conditions of deprivation and/or altering the environment to make motivational “triggers” less aversive • Avoid directly and publicly prompting Mitch to participate orally; instead, collaborate with Mitch to identify an unobtrusive gestural cue for teachers to deliver to prompt participation • Provide options for alternative, nonvocal participation in class • Schedule regular opportunities to take a break from class to engage in self-management strategies (e.g., walk to the water fountain and back while practicing deep breathing)

Sources of Reinforcement (Discriminative Stimuli): Identify strategies to modify or eliminate persons, items, and/or locations that signal the availability of reinforcing consequences for interfering behaviors • Pair teachers with positive reinforcement, while discontinuing the practice of sending Mitch to the principal’s office for disruptive behavior • Assign a seat away from the classroom exit
--

Interfering Behavior(s): Identify the behavior(s) that interfere with functioning • Vocal opposition • Leaving the area	Replacement Behavior(s): Identify logical, socially appropriate replacement behavior(s) • Active class participation • Appropriate requests for a break/permission to leave the classroom
--	--

Reinforcing Consequence(s): Identify strategies for minimizing reinforcement of the interfering behavior • Discontinue the practice of sending Mitch to the principal’s office following interfering behavior	Reinforcing Consequence(s): Identify strategies for maximizing reinforcement of the replacement behavior • Develop and implement a behavioral contract with Mitch to establish goals for active class participation (including reasonable nonvocal ways to participate) and reinforcement options for meeting those goals • Consistently and immediately honor all appropriate requests to take a break or leave the classroom, and allow 5 minutes per break ◦ Gradually fade over time
---	--

FIGURE 11.2. Behavior-Analytic Problem Solving: Function-Focused Intervention planning tool for Mitch. Used with permission from the Margaret Murphy Centers for Children.

A Word or Two about Time-Out (Or . . . Time to Set the Time-Out Record Straight)

Time-out is one of the most overused and misunderstood reactive procedures for addressing interfering behaviors. Technically speaking, the term is “time-out from reinforcement.” Time-out from reinforcement is a punishment procedure designed to reduce occurrences of interfering behavior. The procedure involves removing a student’s access to positive reinforcers for brief periods of time, contingent on occurrences of interfering behavior. This may be accomplished by briefly removing the student’s access to preferred items, activities, and/or social interactions (inclusionary time-out) or by briefly removing the student from a reinforcing environment (exclusionary time-out).

When implemented correctly and consistently, time-out from reinforcement is effective for reducing and eliminating interfering behaviors. However, when a student engages in interfering behavior maintained by negative reinforcement (i.e., avoidance or escape from nonpreferred environments, activities, or social interactions), the use of time-out procedures is likely to strengthen (not weaken) occurrences of interfering behavior. In other words, depending on the function(s) of a student’s interfering behavior, the use of time-out procedures may be contraindicated.

You may ask, “How can that be? Time-out is a form of punishment. How can punishment reinforce behavior?” In behavior analysis, there’s an old saying that goes something like this: *Time-out from reinforcement is effective only to the degree that the time-in environment is reinforcing.* When the time-in environment (e.g., the classroom setting) is not reinforcing and somewhat aversive, the use of time-out is equivalent to providing escape from a nonpreferred situation (Aah . . . relief!). In this situation, the time-out procedures intended to punish interfering behavior may actually reinforce interfering behavior. When we encounter misapplications of time-out like this, we generally use another old saying from the behavior analysis archives that goes something like this: *“Stop using time-out!”*

SUMMARY

Years ago, Dan Reschly made a statement that went something like this: *The only good assessment is the one that results in effective intervention.* This premise has been a driving force throughout our professional careers. From our perspective, FBAs that result only in a conceptualization of the “problem” are inadequate. *Effective* FBAs lead to the selection and implementation of comprehensive, individually tailored behavior intervention plans that include empirically validated strategies for (1) reducing occurrences of interfering behavior and (2) increasing socially appropriate behavior.

In this chapter, we did not attempt to provide a comprehensive review of evidence-based behavioral intervention strategies. The broad topic of behavioral interventions is far beyond the scope of a single chapter! Instead, our goal was to illustrate a conceptually sound approach to developing comprehensive behavior intervention plans that are “function focused” and informed by FBA results. By developing intervention plans that explicitly address each individual, antecedent, and consequence variable that contributes to occurrences of interfering behavior, you will be far more successful in achieving socially significant outcomes for your students.

In conclusion, we advocate using only interventions that are both function based and evidence based, and we leave you with a final reflection on our responsibility as interventionists . . .

Trick or Treatment

Throughout this book we have emphasized that evaluators have a professional and ethical responsibility to obtain informed consent prior to conducting an FBA. We also assert that practitioners have an equal responsibility to obtain *informed consent prior to implementing interventions*.

In our years of providing school psychology services, we have found that our professional colleagues sometimes disregard science and insist on recommending and implementing interventions that lack empirical evidence. We believe that school teams have a responsibility to be honest and transparent about the evidence base for recommended interventions and thoroughly explain the likelihood that recommended interventions will yield positive outcomes for referred students. Following are (somewhat tongue-in-cheek) examples of the conversations practitioners may have with caregivers before implementing interventions.

EVIDENCE-BASED INTERVENTION

“Good morning. We have completed a comprehensive FBA, and based on the results of this assessment, we recommend a strategy called functional communication training (FCT) to increase Michelle’s communication skills and decrease occurrences of interfering behaviors. Hundreds of research studies have demonstrated the effectiveness of FCT for similar students. Furthermore, during our FBA, we conducted a brief treatment analysis to ‘test-drive’ this strategy with Michelle. We taught her to request breaks or assistance from her teacher, honored these requests by giving her a 5-minute break or assistance with the assigned task, and stopped providing breaks and assistance when she engaged in self-injurious behaviors. Results of this treatment analysis showed a significant reduction in self-injury and increases in on-task behavior when we taught Michelle an alternative way to communicate her needs. We are confident that this will be an effective intervention, and we intend to (1) train her teachers to follow the plan accurately and (2) collect progress-monitoring data to ensure that she is responding positively. Do you have any questions?”

NON-EVIDENCE-BASED INTERVENTION

“Good morning. I have not conducted an FBA. Based on my years of clinical experience with students like Michelle, I recommend that we use dill pickle therapy (DPT)* to increase on-task behavior and reduce self-injurious behavior. Although there is no research evidence to support this approach, I am really hoping that it will be helpful. I have not tried it out with Michelle, but I have seen it work with other students and believe it will be beneficial. I must point out that there are evidence-based strategies we could try, and there is a possibility that DPT will make things worse for Michelle. Rest assured, though, I will check in with Michelle regularly and give you my clinical opinion on how she is doing. I hope you will trust my clinical insights and professional opinion. Do you have any questions?”

So, if Michelle was your child, would you go with FCT or DPT (the “trick” of the trade)?

From our perspective, the examples above illustrate the difference between *ethically responsible* and *reckless* treatment.

- *Ethically responsible treatment*: Interventions are evidence based and function based, and progress-monitoring data are collected to ensure they are effective for the individual student.
- *Reckless treatment*: Interventions are developed and evaluated on the basis of (1) poorly designed and executed research, (2) unproven theories, (3) supposition, (4) biased anecdotal observations, (5) personal opinion, or (6) subjective testimonials.

*Although we used DPT as a humorous example, we have participated in far too many team meetings where practitioners have enthusiastically and confidently recommended interventions that have not been empirically validated.

SOLUTIONS

1. Many resources exist to help you determine if a given intervention is evidence based. Share these resources to prevent “trick treatments” from continuing to be perpetrated against vulnerable parents and students.
 - The National Standards Project (2015; www.nationalautismcenter.org/national-standards-project) offers objective reviews of treatment research by an expert panel. Interventions are rated as established (solid research support), promising (mixed or emerging research support), unestablished (no solid research support), or harmful.
 - The What Works Clearinghouse (WWC; <https://ies.ed.gov/ncee/wwc/FWW>) is a resource maintained by the Institute of Education Sciences (a branch of the U.S. Department of Education). It offers reviews of existing research to inform decision making about evidence-based practices in schools.
2. When interventions with inadequate empirical support are recommended, remember our motto: *When in doubt . . . test it out!* We advocate for conducting a quick treatment analysis to determine whether a particular intervention will be effective for a particular student (see Chapter 12).

FORM 11.1

Behavior-Analytic Problem Solving: Function-Focused Intervention

Name: _____ **Date:** _____

Personal Characteristics: Identify strategies to address traits, sensitivities, beliefs, and/or values contributing to interfering behavior	Context: Identify strategies for modifying contexts and/or participation in contexts in which interfering behavior occurs	Skill Delays/Deficits: Identify strategies to mitigate communication, academic, executive, adaptive living, social, and/or self-management skill delays/deficits contributing to interfering behavior

Motivational “Triggers” (Motivating Operations): Identify strategies to weaken (abolish) motivation for interfering behaviors by preventing conditions of deprivation and/or altering the environment to make motivational “triggers” less aversive

Sources of Reinforcement (Discriminative Stimuli): Identify strategies to modify or eliminate persons, items, and/or locations that signal the availability of reinforcing consequences for interfering behaviors

Interfering Behavior(s): Identify the behavior(s) that interfere with functioning	Replacement Behavior(s): Identify logical, socially appropriate replacement behavior(s)

Reinforcing Consequence(s): Identify strategies for minimizing reinforcement of the interfering behavior	Reinforcing Consequence(s): Identify strategies for maximizing reinforcement of the replacement behavior

Used with permission from the Margaret Murphy Centers for Children.

From *Conducting School-Based Functional Behavioral Assessments, Third Edition*, by Mark W. Steege, Jamie L. Pratt, Garry Wickerd, Richard Guare, and T. Stuart Watson. Copyright © 2019 The Guilford Press. Permission to photocopy this material is granted to purchasers of this book for personal use or use with students (see copyright page for details). Purchasers can download additional copies of this material (see the box at the end of the table of contents).